

Diagram illustrating the SWD (Serial Wire Debug) interface connections for the J4 module. The connections are as follows:

- SWCLK (Serial Wire Clock) is connected to pin 1.
- SWDIO (Serial Wire Data In/Out) is connected to pin 2.
- A common ground connection is connected to pin 3.

The module is labeled J4 SWD.

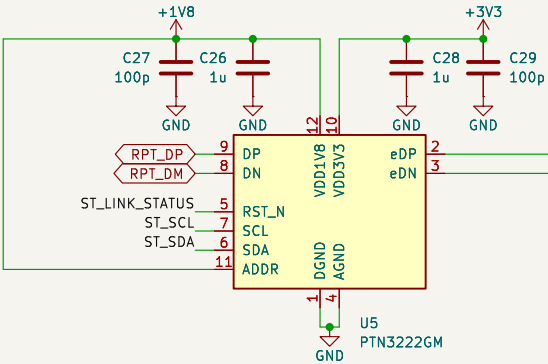
J2 LEFT

Pin	Signal
1	+1V8
2	GND
3	TUN_G0
4	TUN_G1
5	CFG_SDA
6	CFG_SCL
7	GP21
8	GP22
9	GP23
10	GP24
11	GP26
12	GP27
13	GP28
14	GP29
15	+3V3
16	VBUS

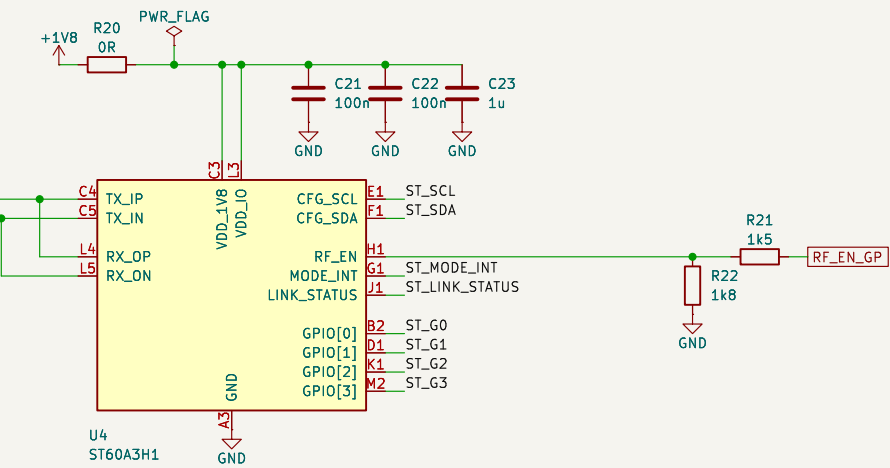
J3 RIGHT

Pin	Signal
1	ST_INT
2	ST_LINK
3	TUN_G2
4	TUN_G3
5	RF_EN_GP
6	GP7
7	GP6
8	GP5
9	GP4
10	GP3
11	GP2
12	GP1
13	GP0
14	GND
15	+3V3
16	VBUS

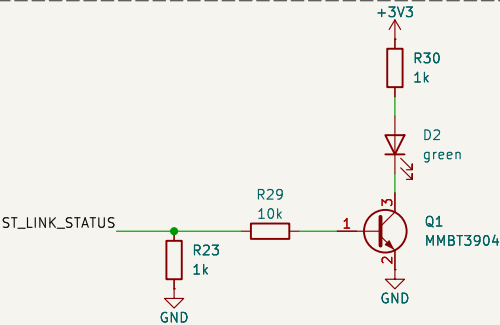
eUSB2 repeater



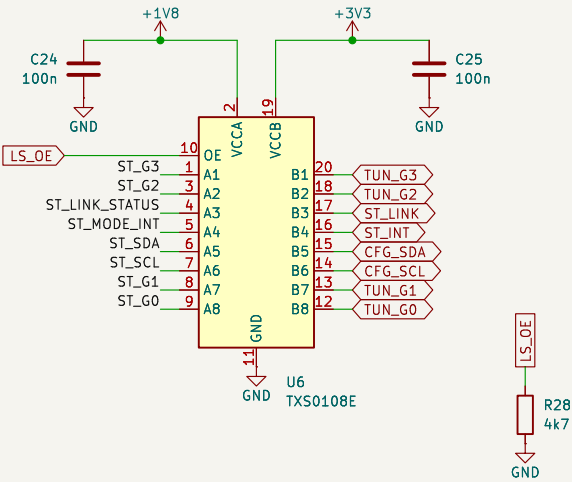
ST60A3H1



Link LED

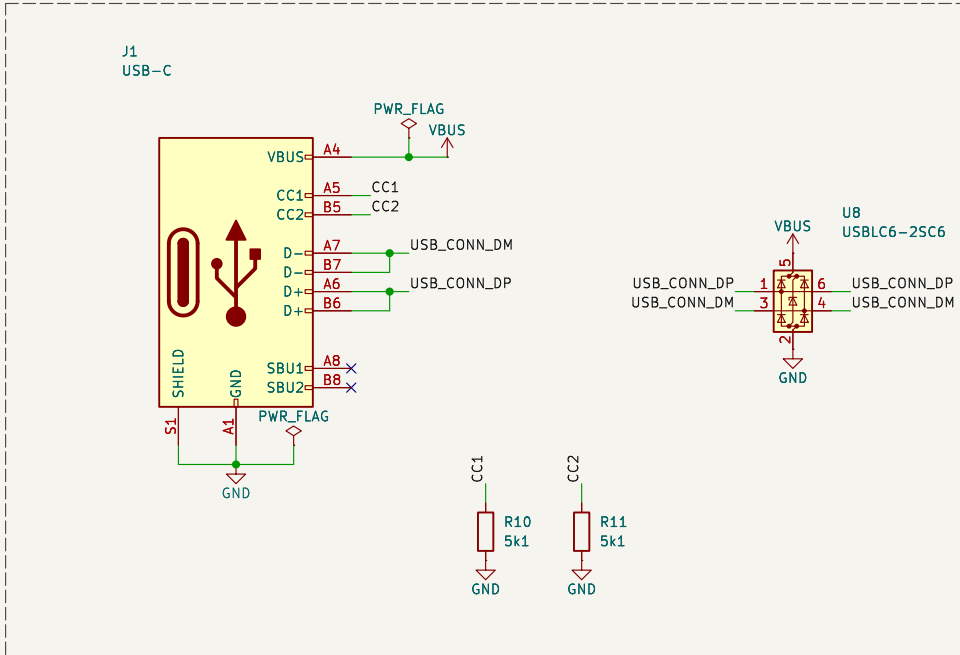


1.8 V – 3.3 V translator

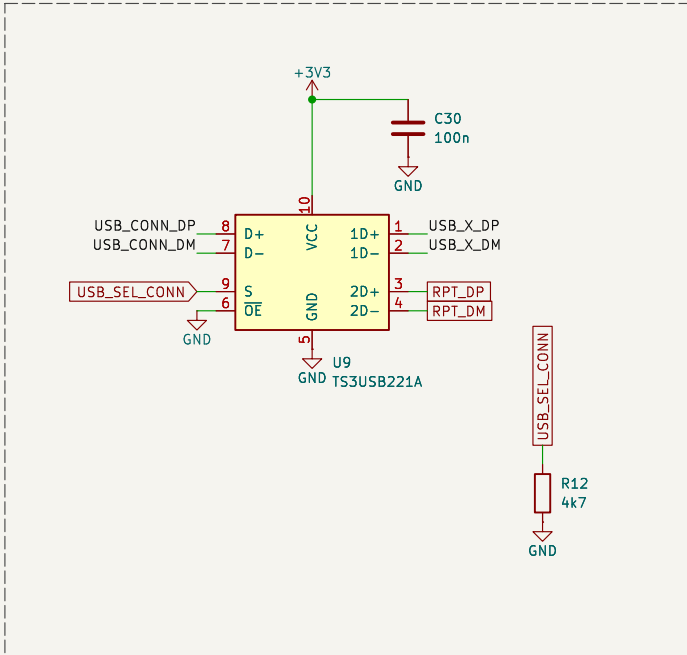


After ST's X-NUCLEO-60K1A1 (B2379A): the ST60A3H1 on +1V8 through a 0 R link, the PTN3222 on its eUSB2 pair and held in reset by LINK_STATUS, an eight-channel auto-direction translator to 3.3 V (TXS0108E here, ST2378E there), RF_EN by divider, the link LED off an NPN.
The translator is off (LS_OE low) until the RP2350 enables it, so the ST60 powers up with only its own straps on its pins. No pull-ups are fitted on the configuration bus or the tunnel pins: ST leaves its own unfitted. A tunnelled I2C bus wants pull-ups on J2/J3, at 3.3 V.
Configuration bus: ST60A3H1 at 0x60, PTN3222 at 0x43 (ADDR on +1V8).

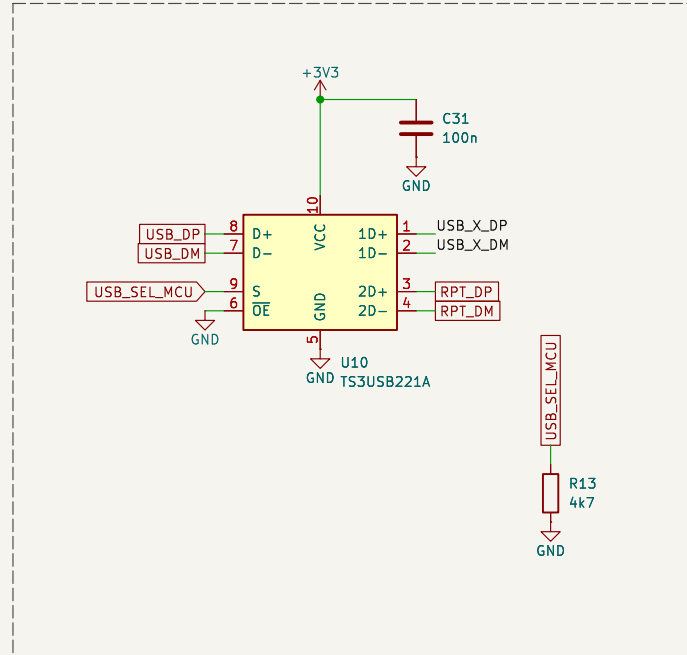
USB-C, device



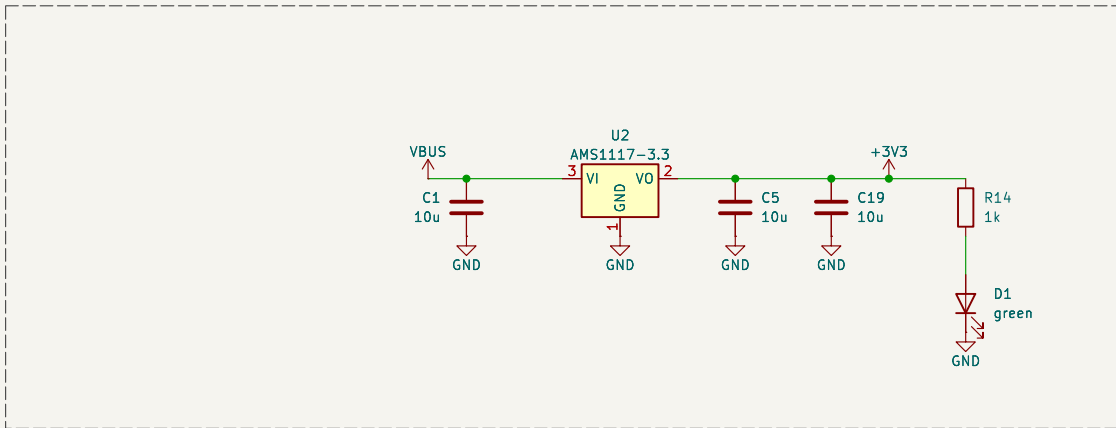
Switch: connector side



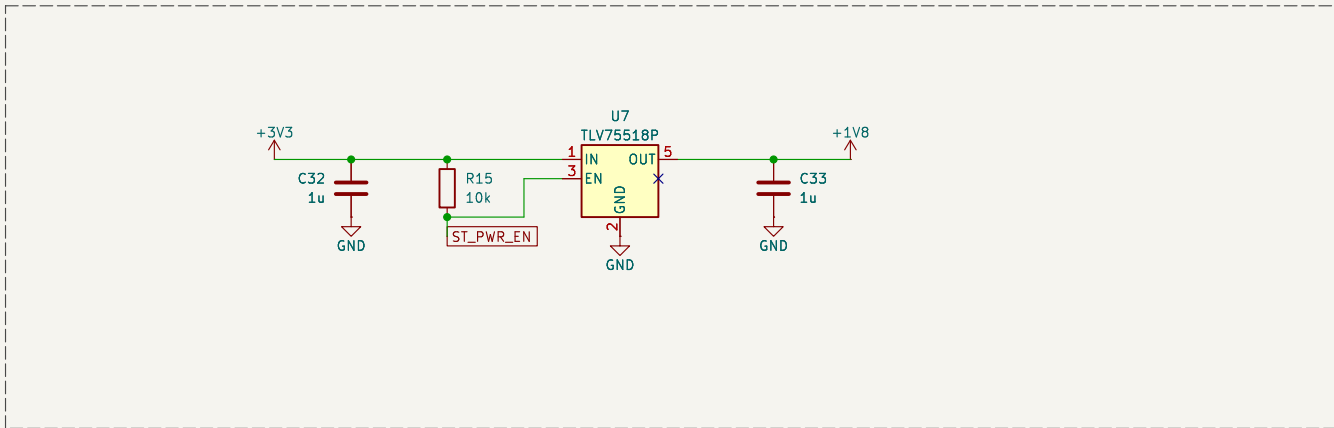
Switch: RP2350 side



3.3 V



1.8 V, switchable



Three things want the USB pair: the connector, the RP2350, the USB2 repeater. The two switches join any two through the node USB_X.

USB_SEL_CONN low, USB_SEL_MCU low: connector to RP2350 -- the board as it powers up, both selects pulled down.

CONN high: connector to repeater -- a host's USB goes over the air.

MCU high: RP2350 to repeater -- the RP2350 is the device at the far end.

Both high joins all three and is not a state to use.

ST_PWR_EN low turns the 1.8 V regulator off; it discharges its output, which is a power-on reset for the ST60A3H1.